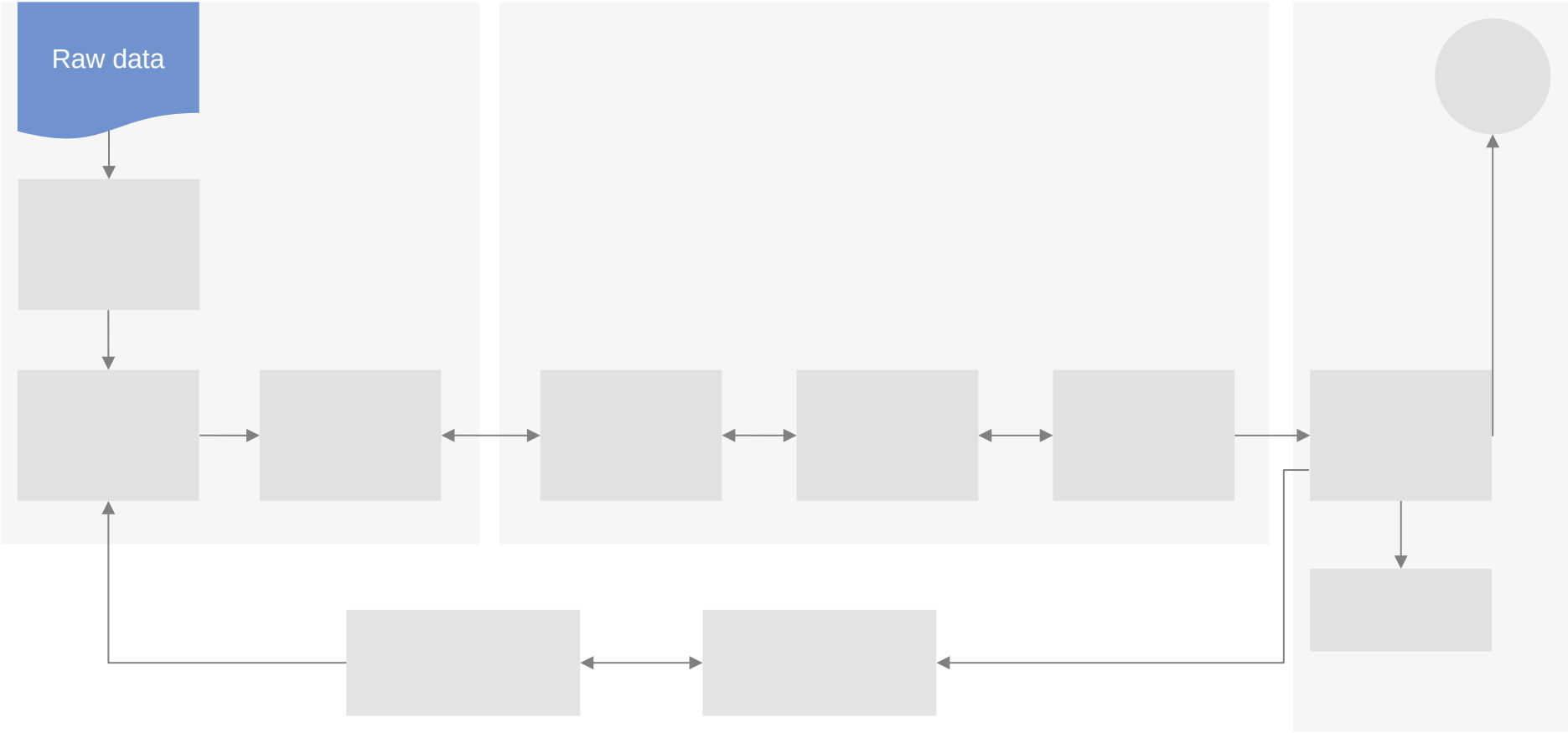
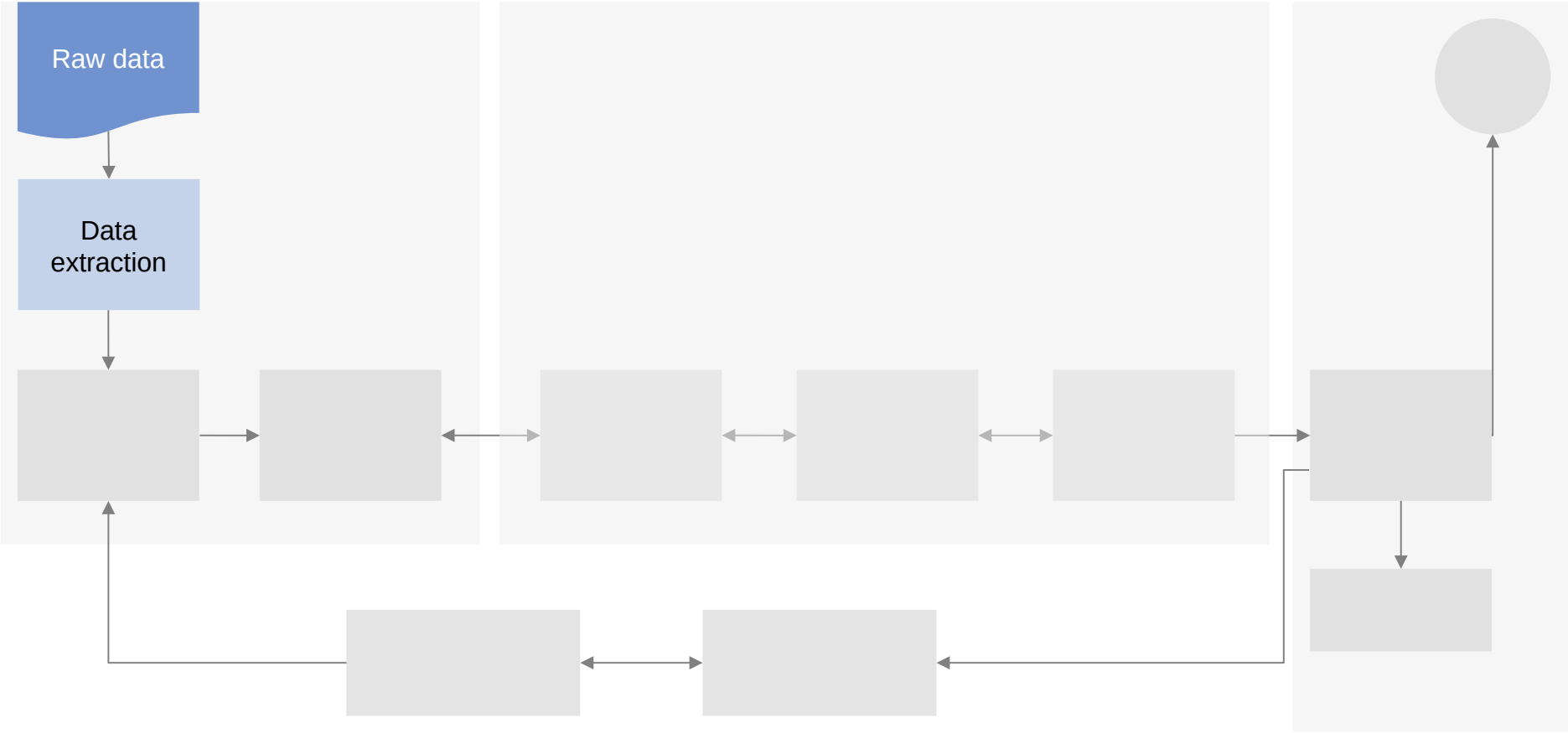
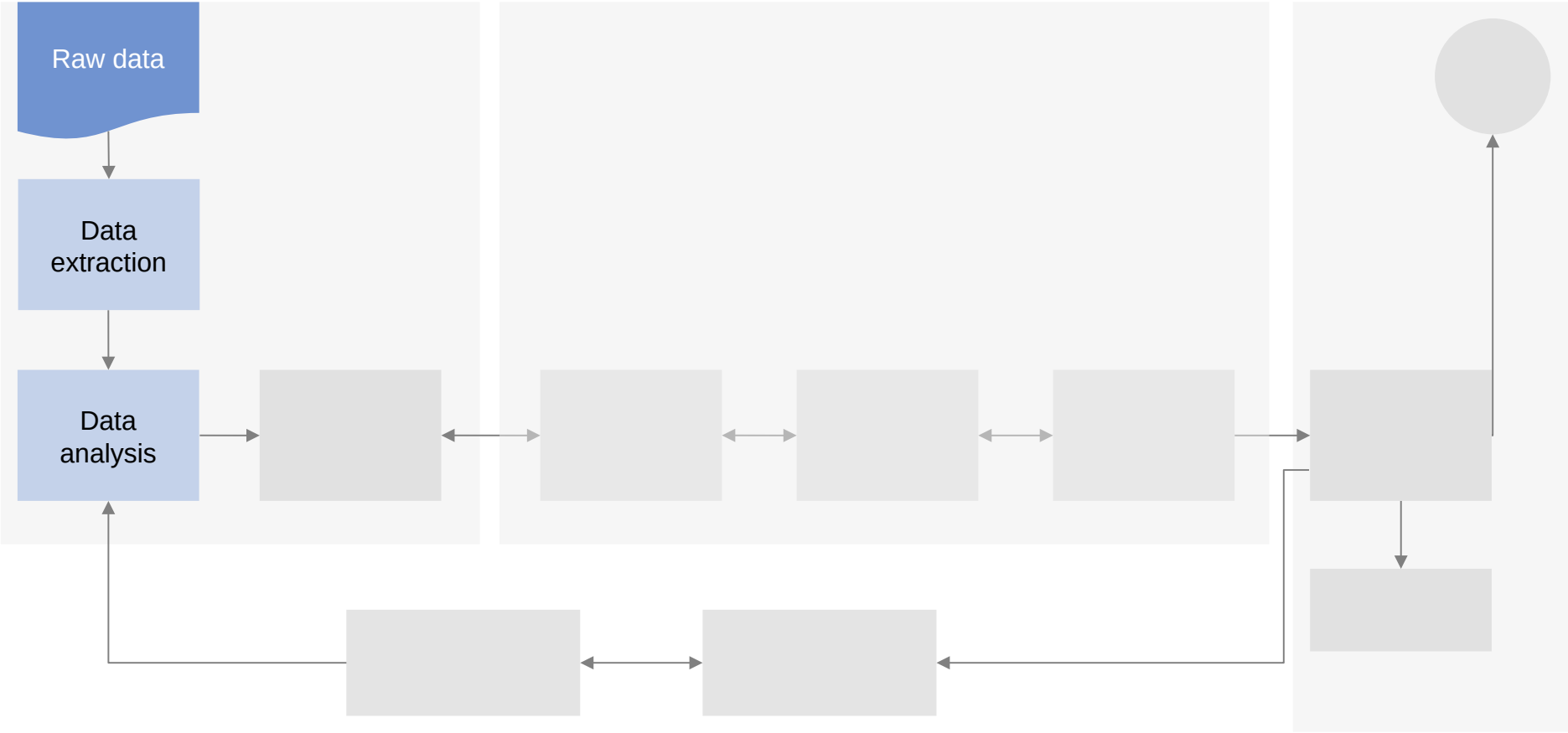
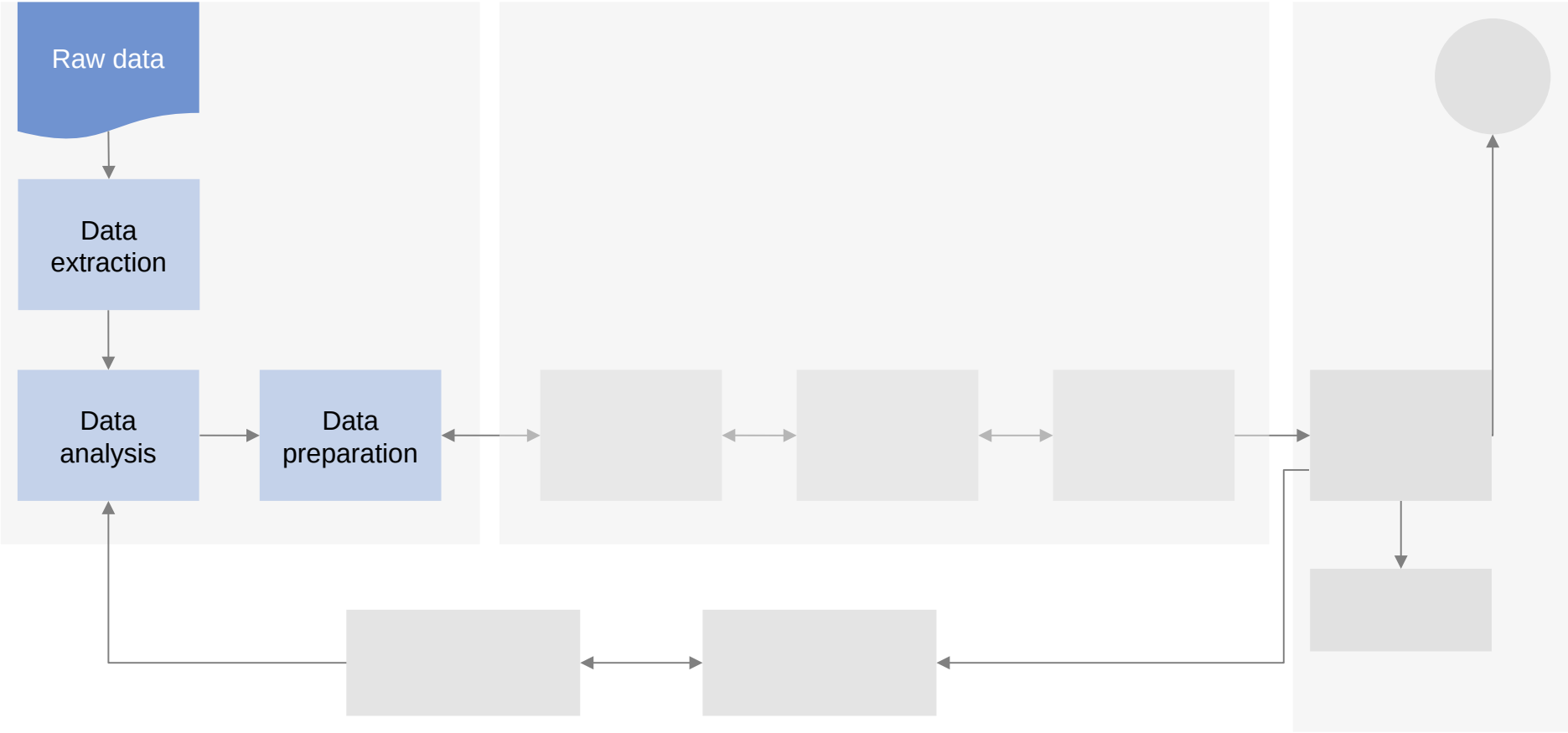
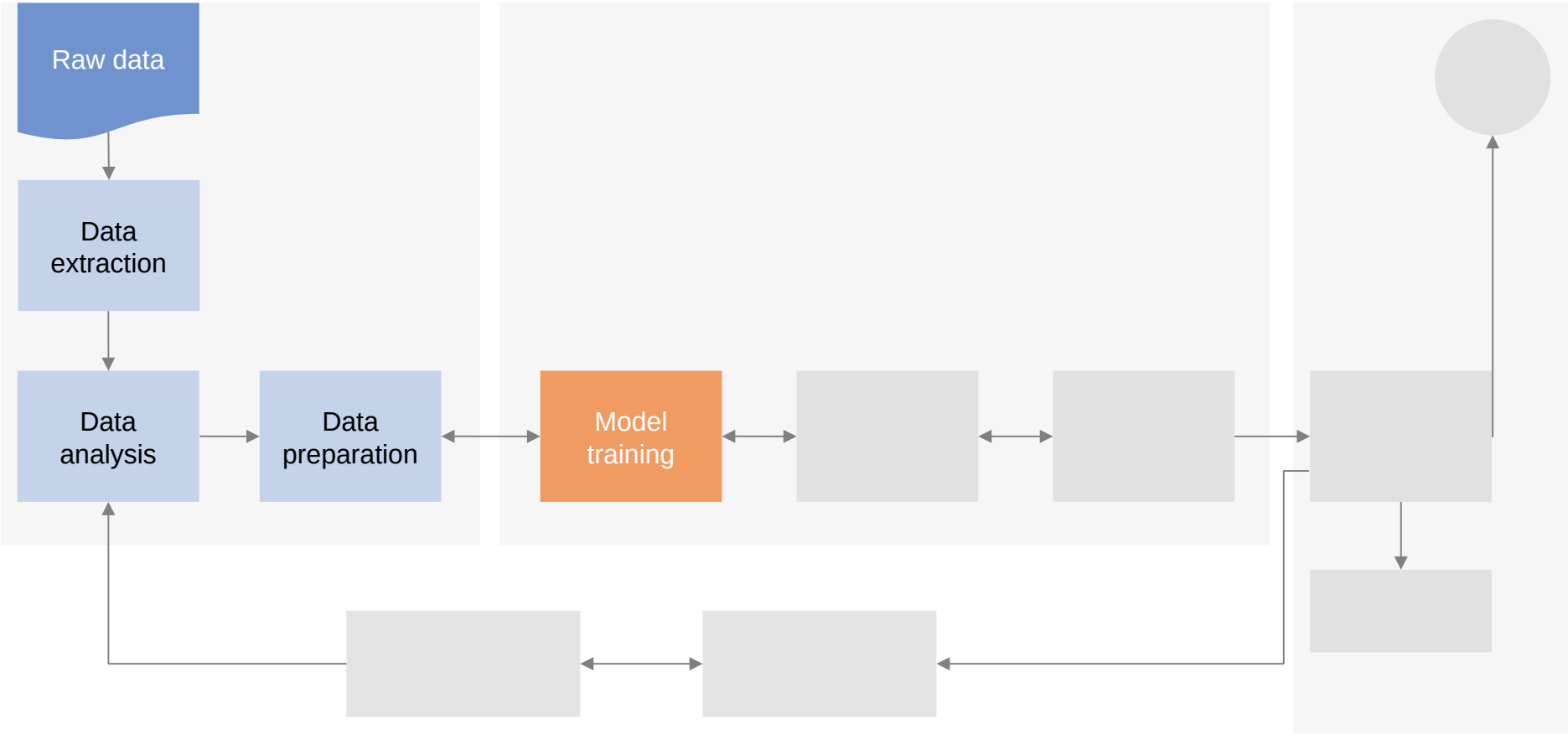
ML Model Lifecycle

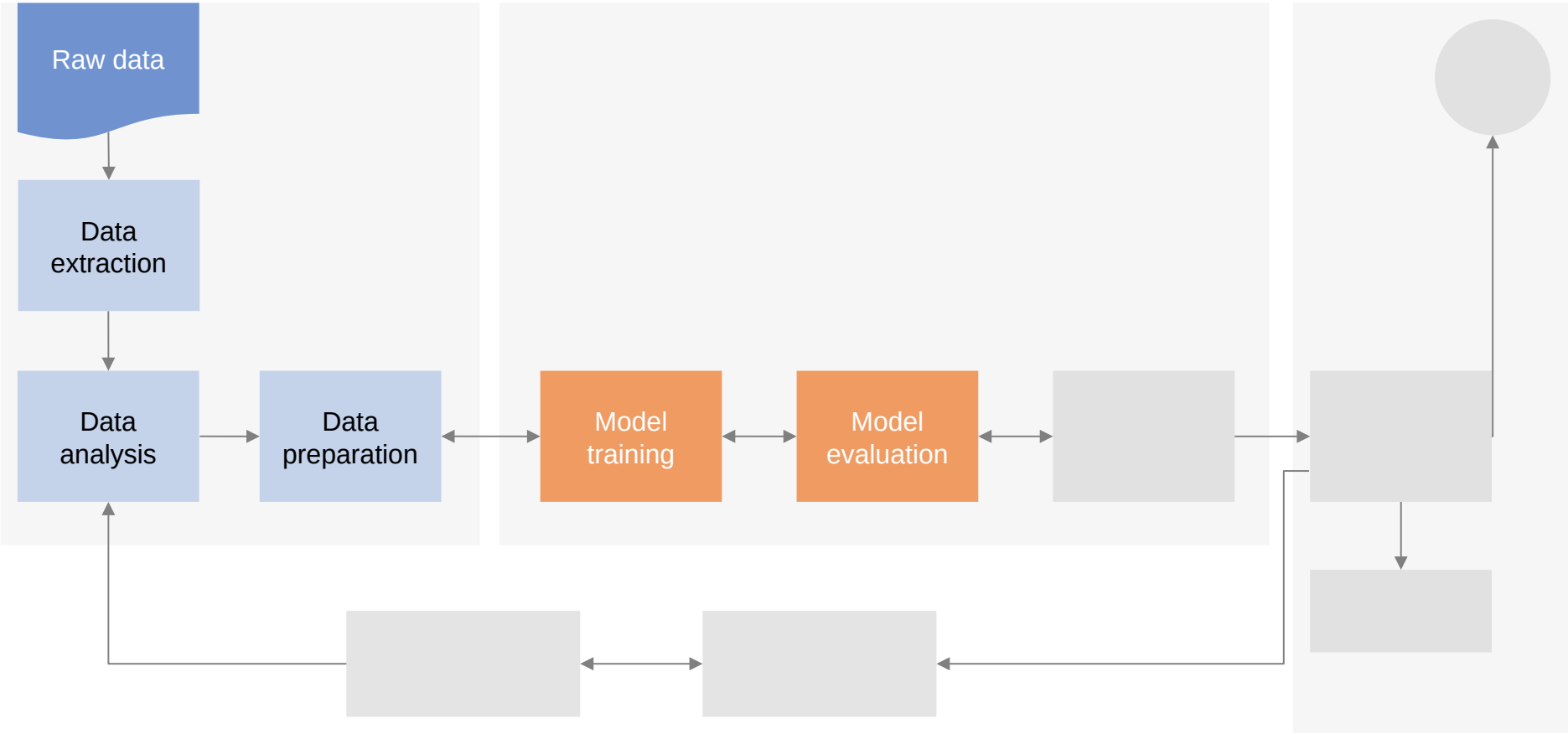

Raw data

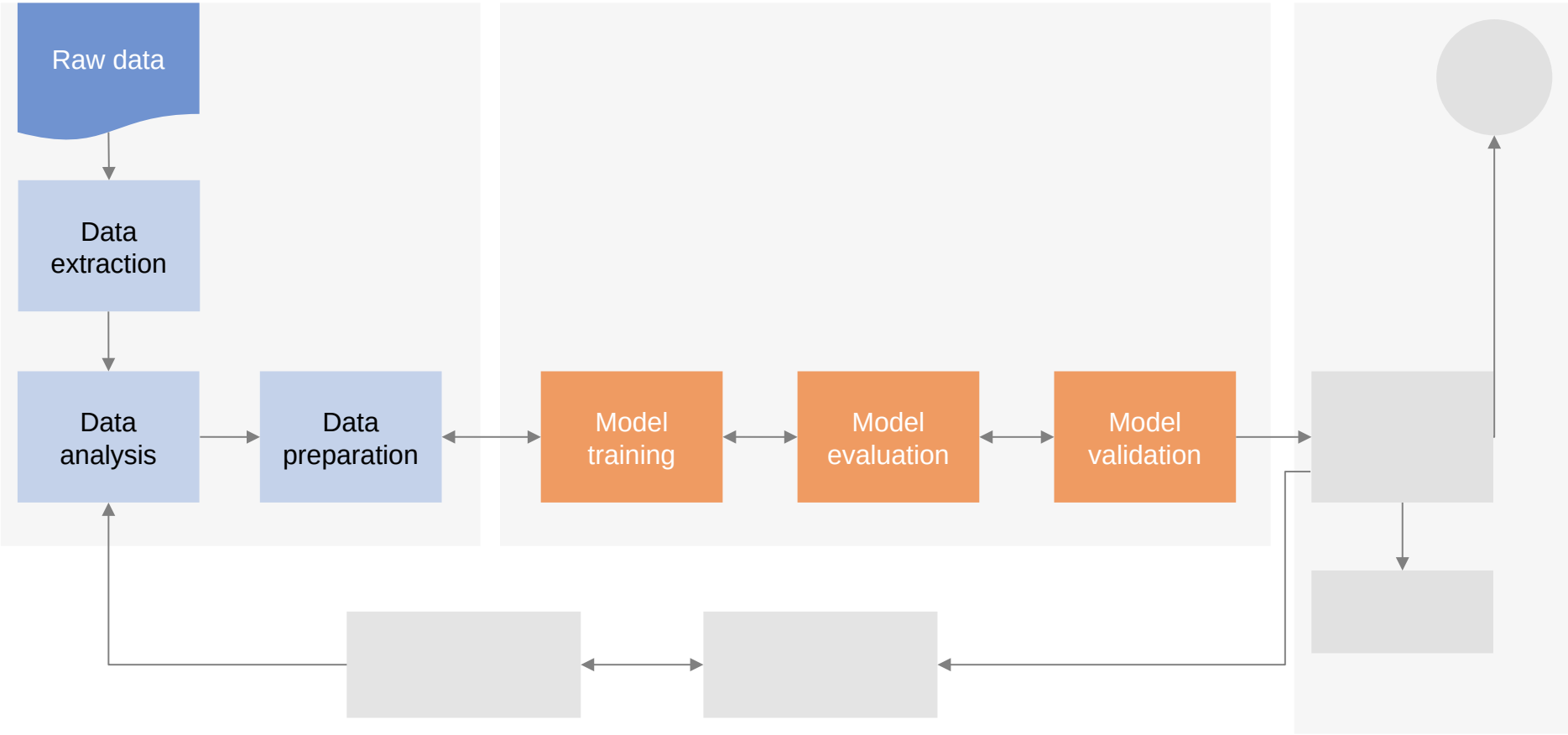

Raw data
Data extraction

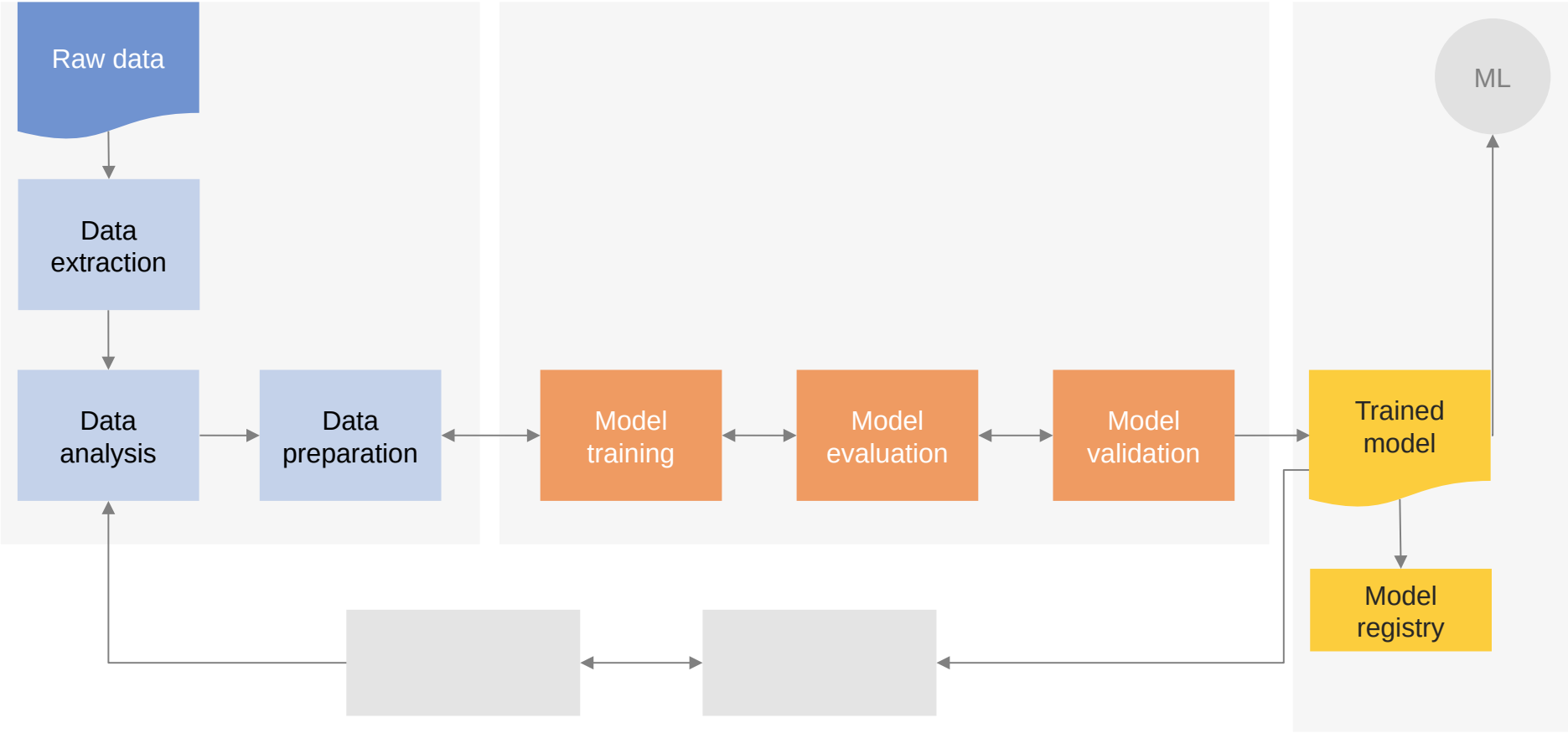

Raw data
Data extraction
Data analysis

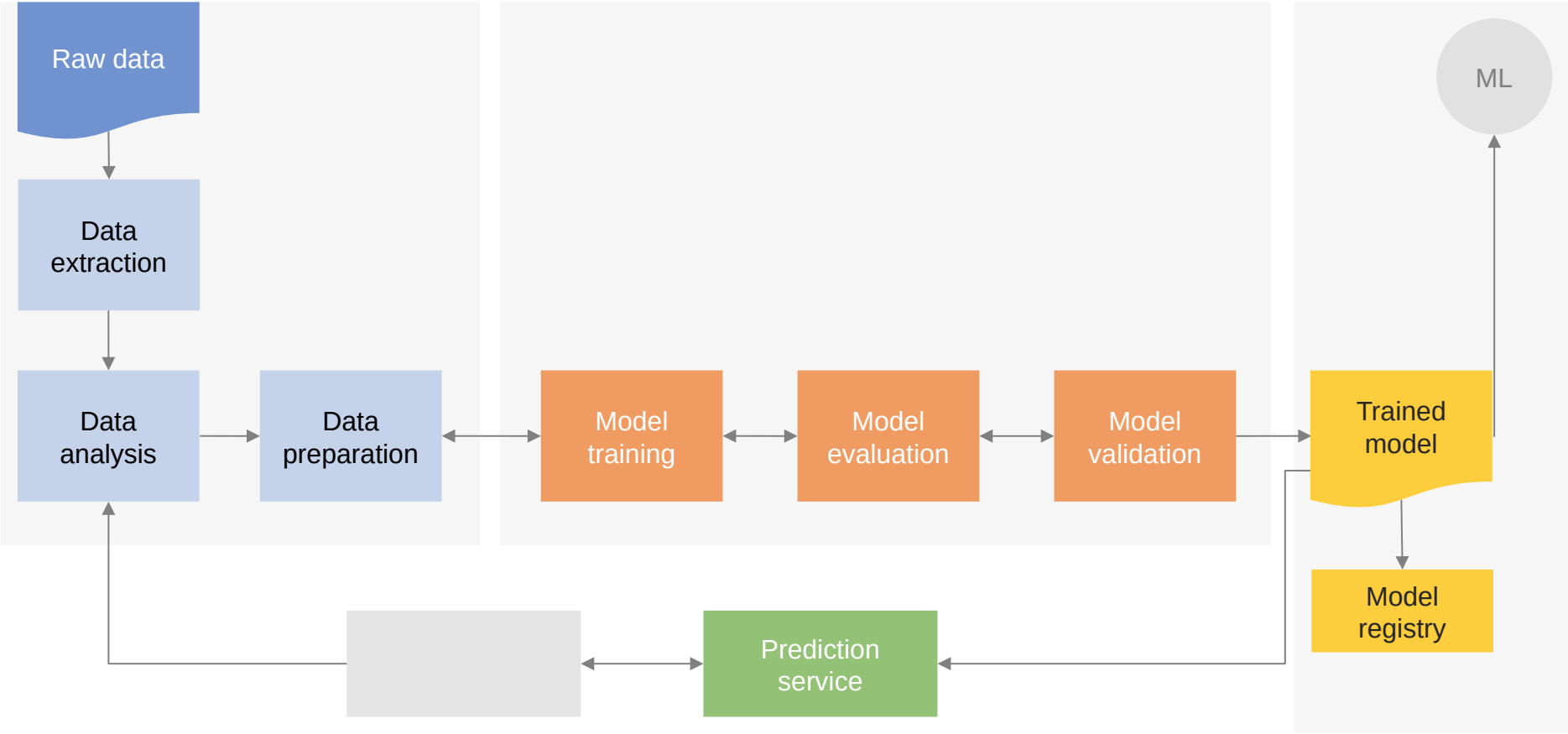

Raw data
Data extraction
Data analysis
Data preparation

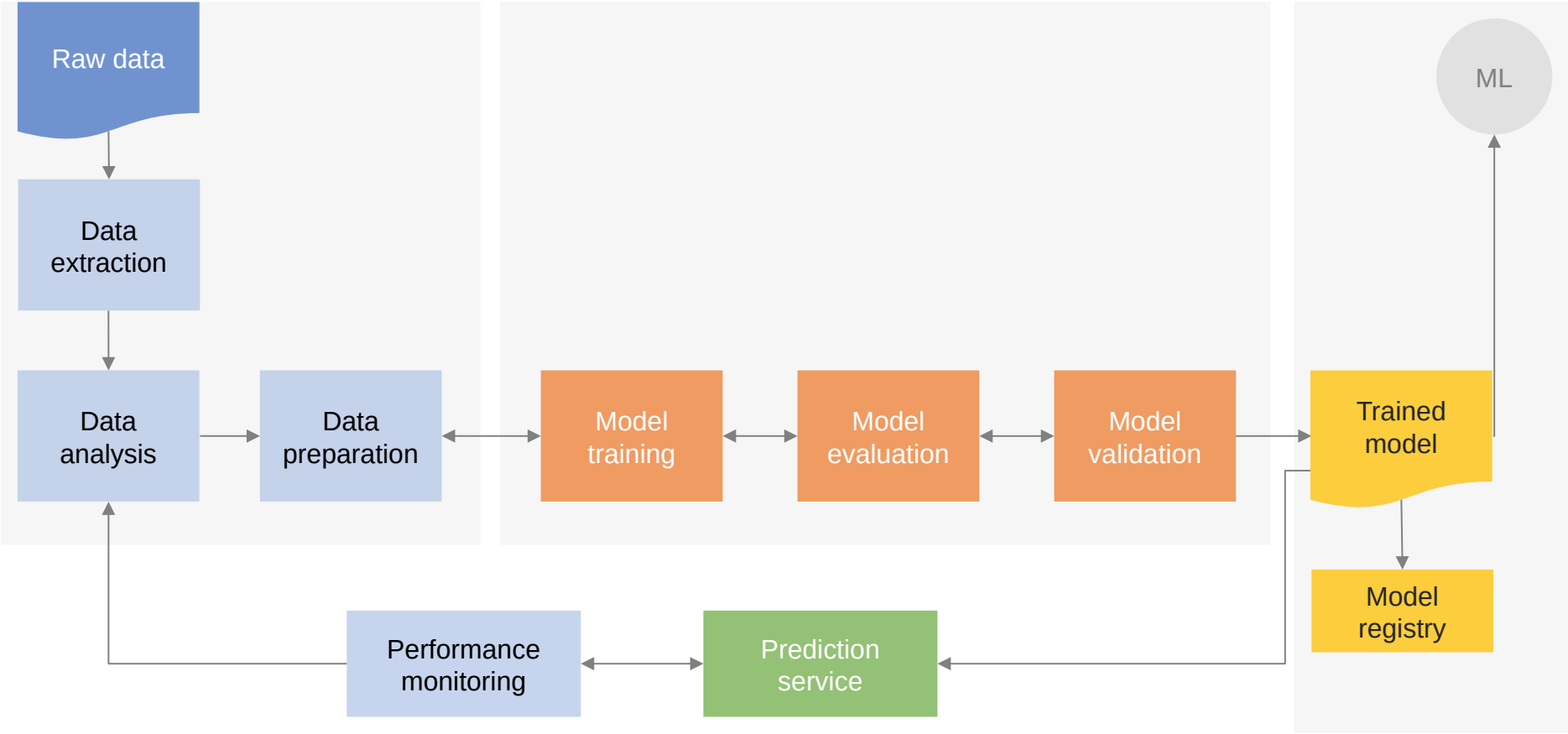

Raw data
Data extraction
Data analysis
Data preparation
Model training


Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation

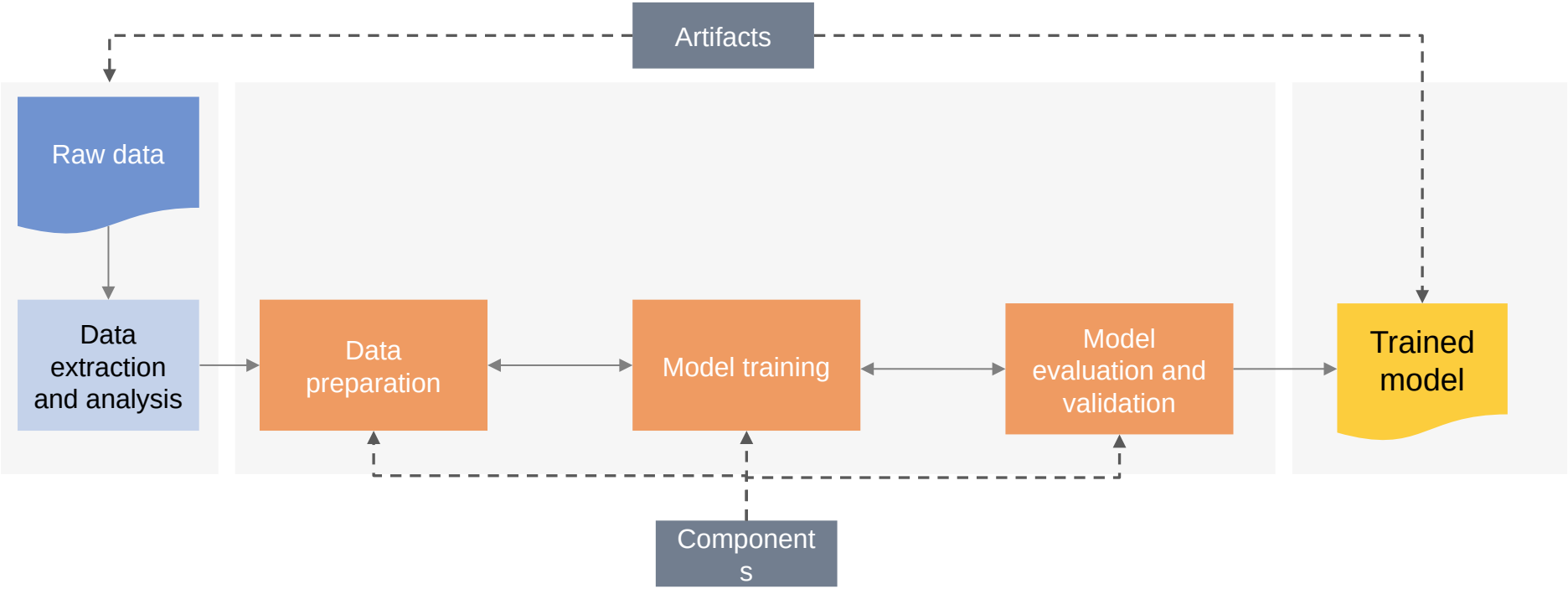

Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation
Model validation


Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation
Model validation
Trained model
ML
Model registry


Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation
Model validation
Trained model
ML
Model registry
Prediction service


Raw data
Data extraction
Data analysis
Data preparation
Model training
Model evaluation
Model validation
Trained model
ML
Model registry
Performance monitoring
Prediction service

ML Pipeline consists of artifacts and components



Components „do” things

Remember WBS?

Work Breakdown Structure

Model training

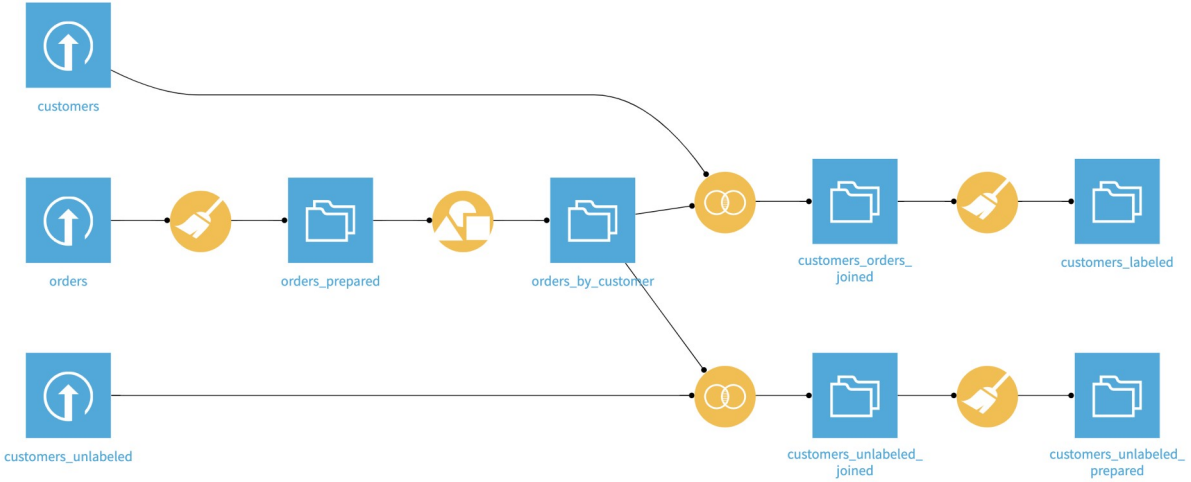
Artifacts are the „things”

Remember PBS?

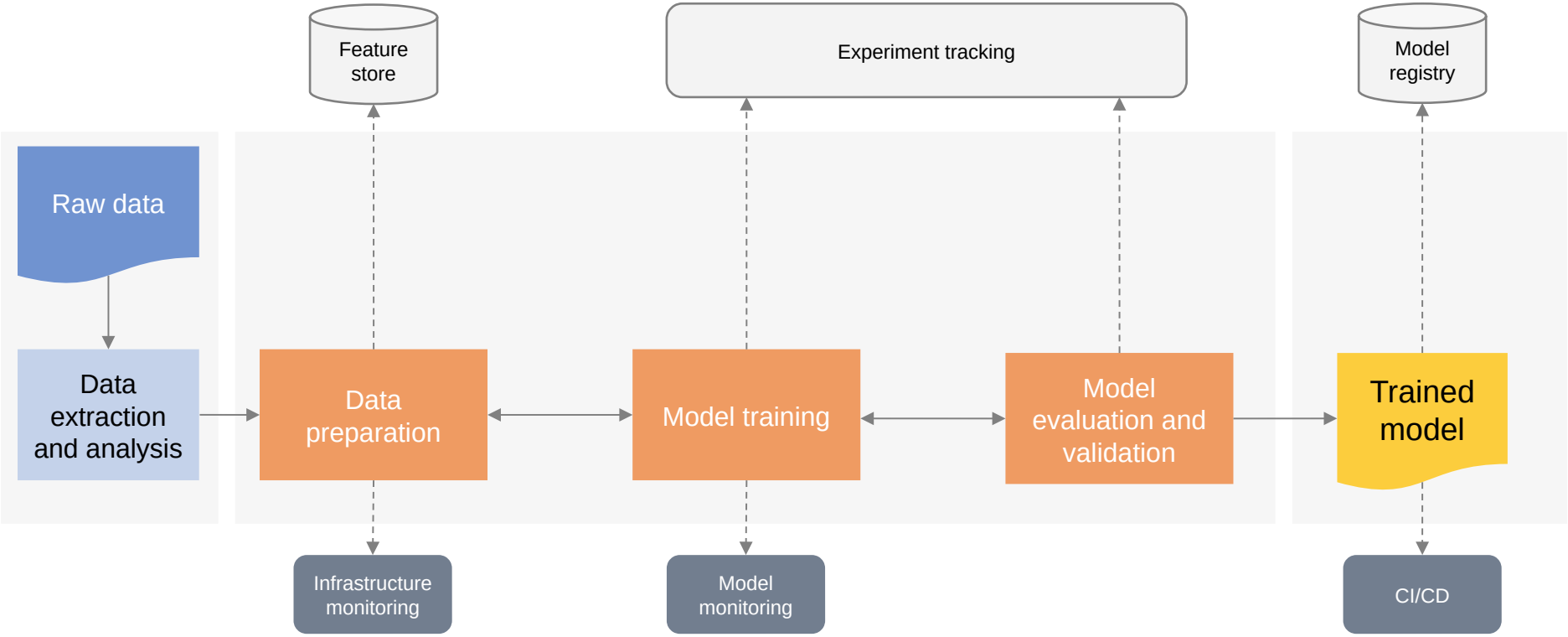
Product Breakdown Structure

Trained model

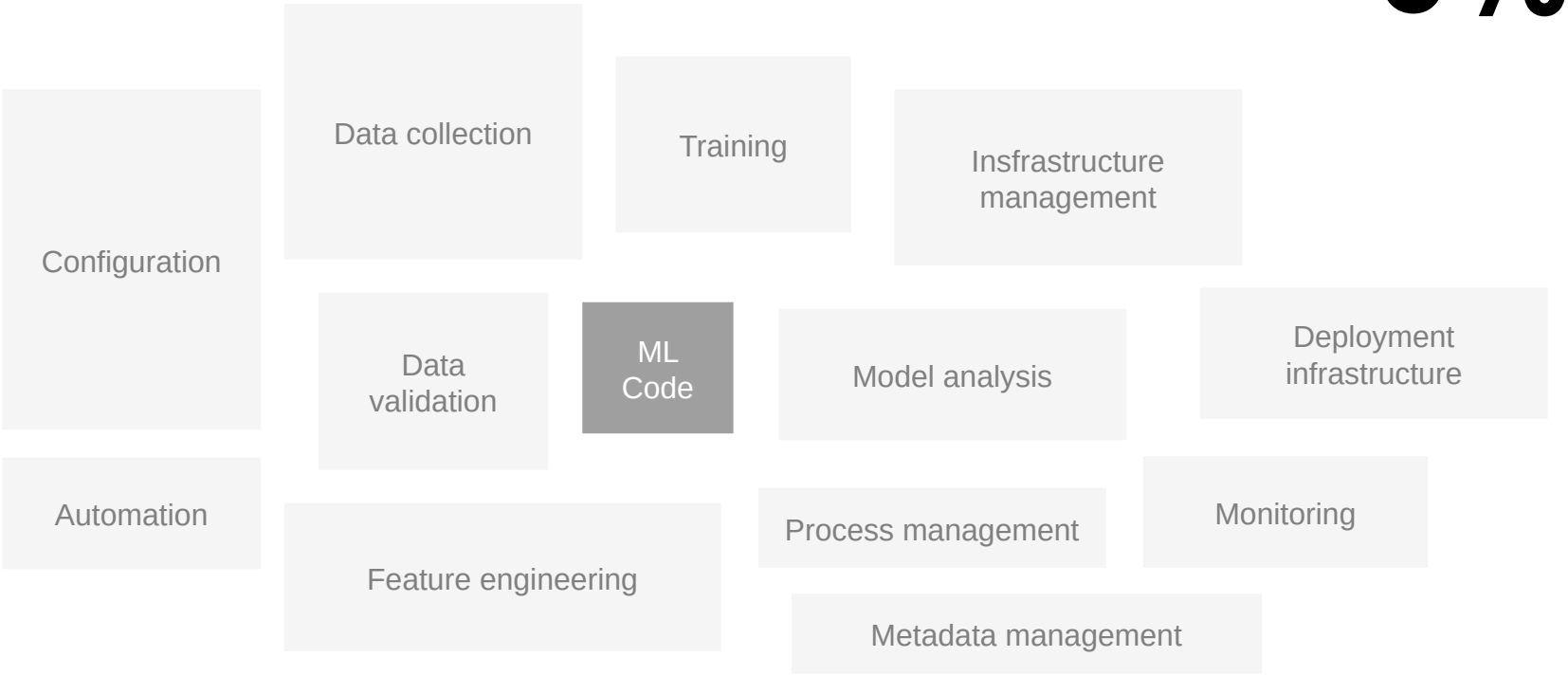
That’s often visualised in an ML pipeline

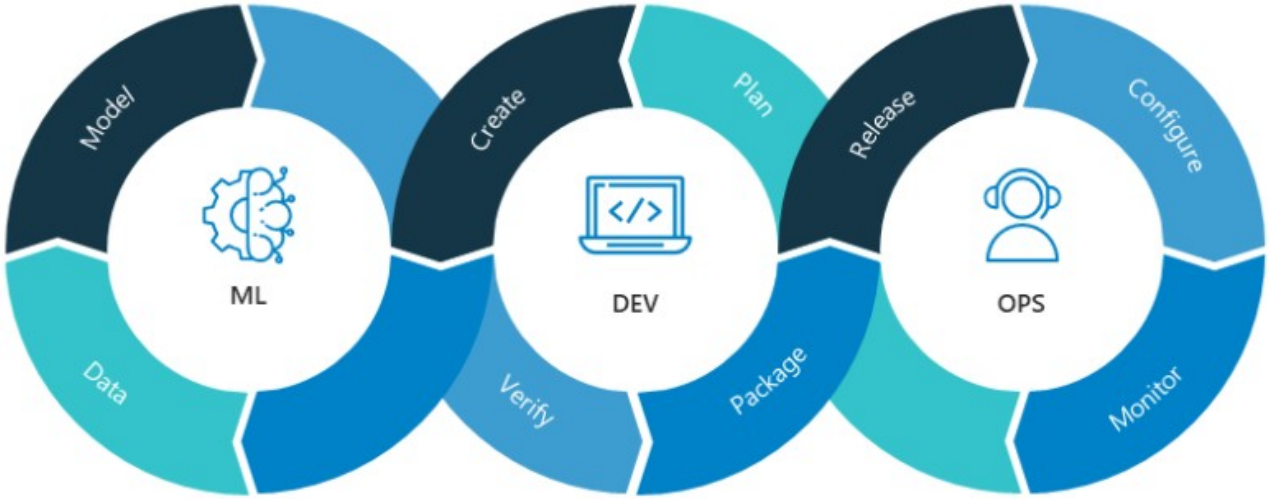


The whole proces is supported by repositories and CI/CD systems



5%


Configuration
Data collection
Training
Insfrastructure management
Data validation
ML Code
Model analysis
Deployment infrastructure
Automation
Feature engineering
Process management
Monitoring
Metadata management


Model
Data
ML
Create
Verify
DEV
Plan
Package
Release
Monitor
OPS
Configure

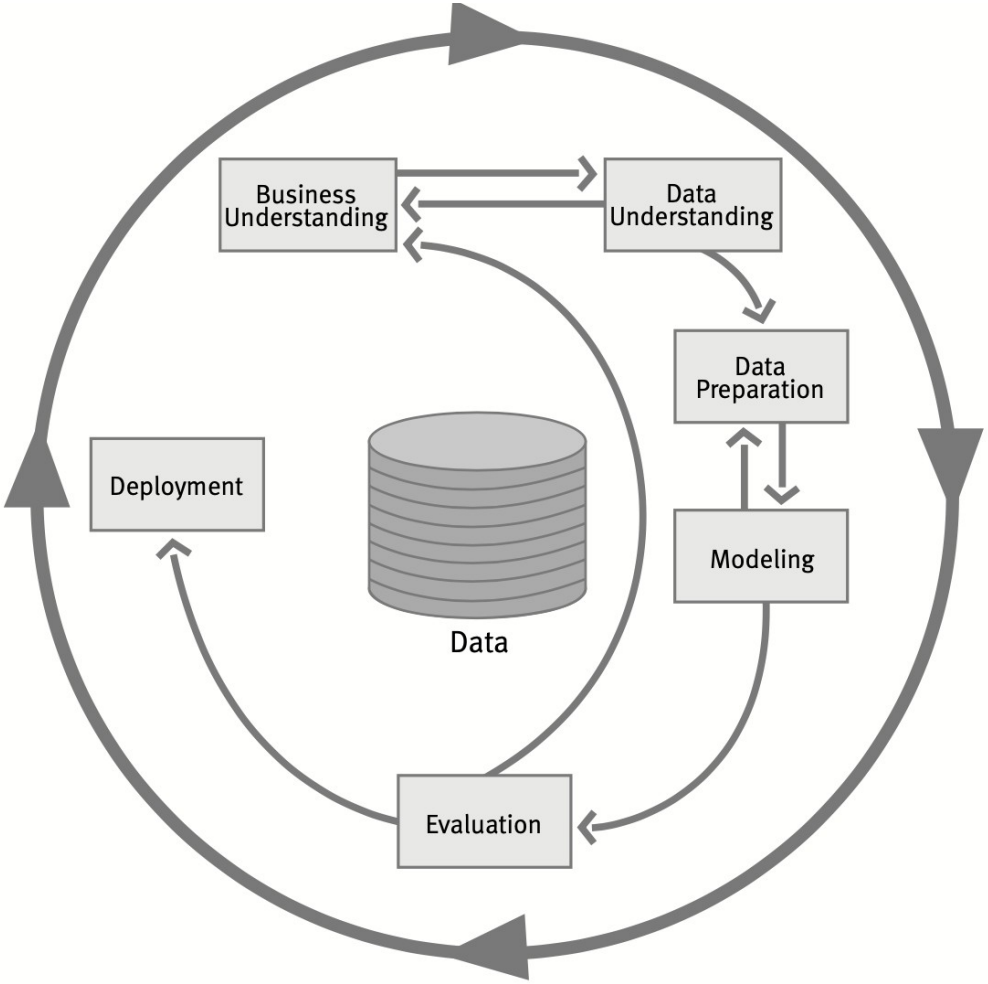
Dive In

Model Development

Develo
p
01
Deplo
y
02
Montio
r
03

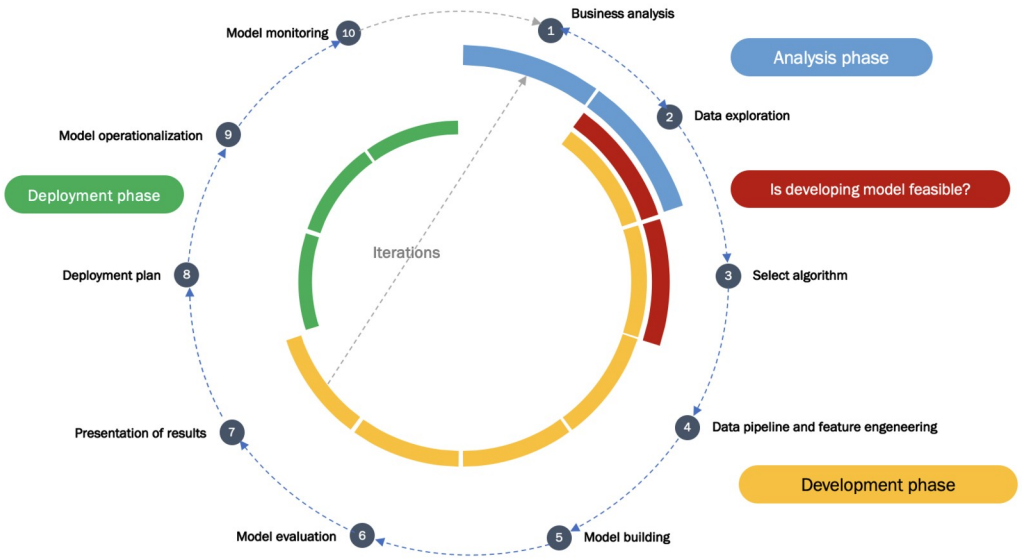
CRISP-DM

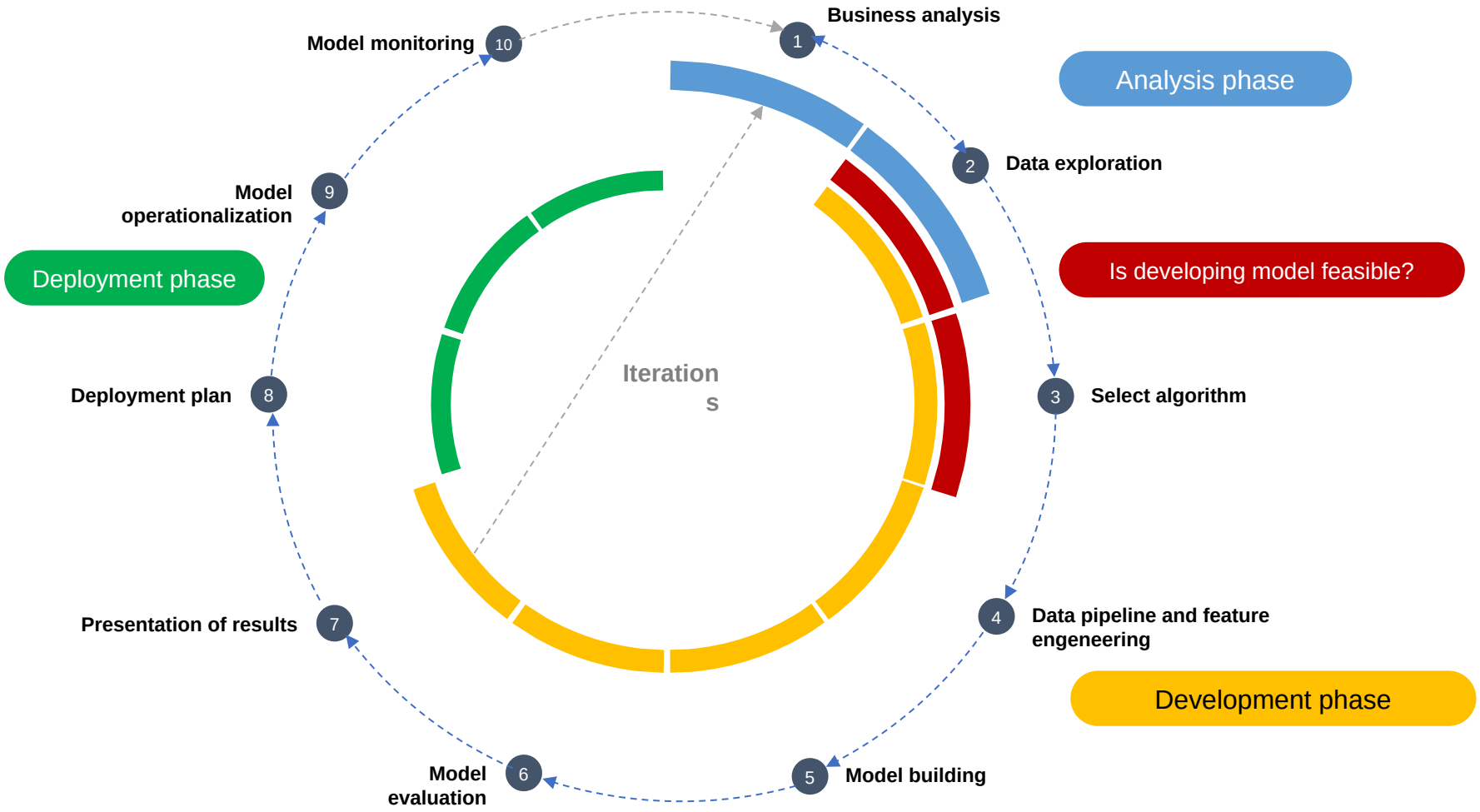
> 20 years old, but works ☺

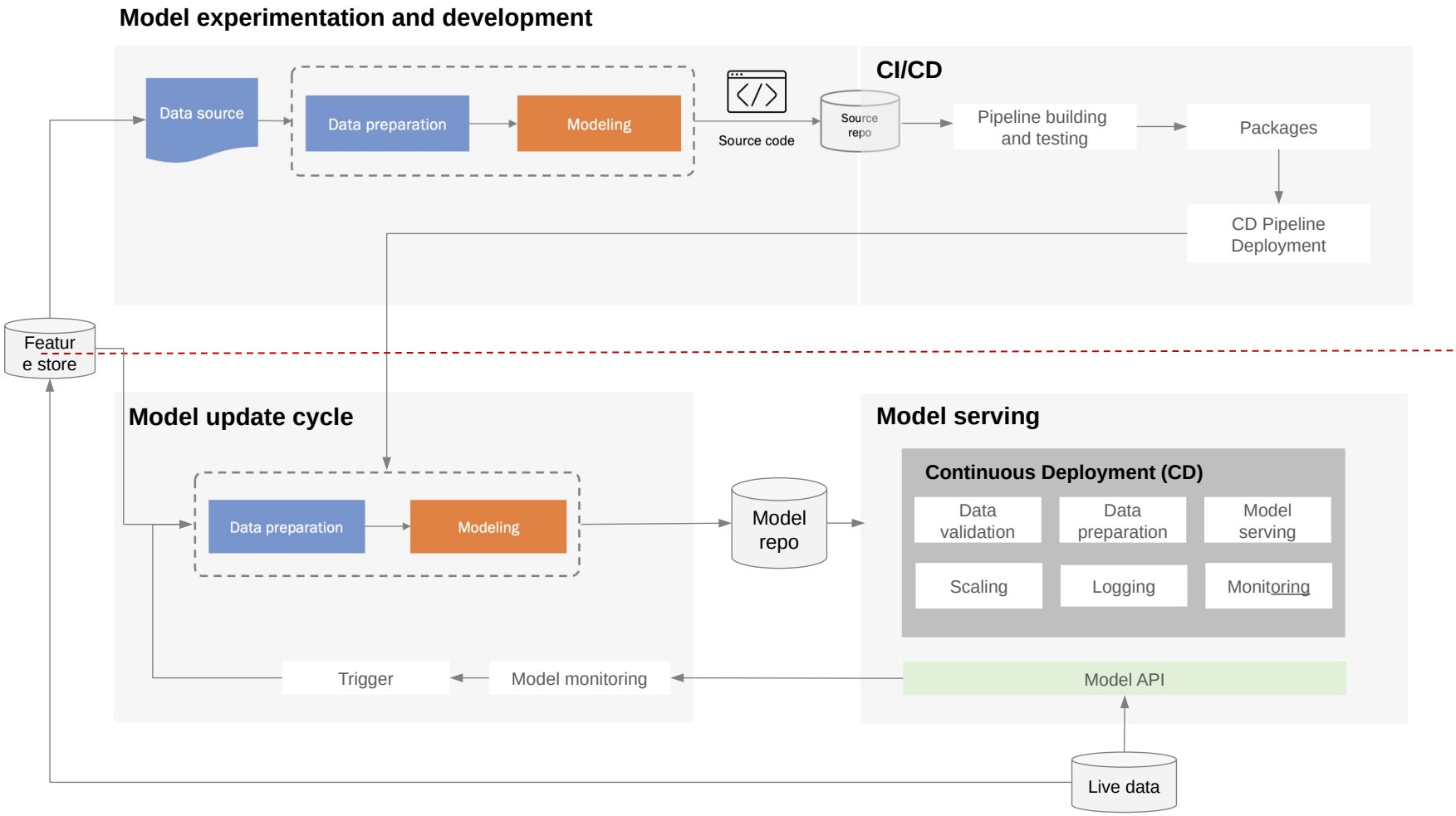


Business Understanding	Data Understanding	Data Preparation	Modeling	Evaluation	Deployment
Determine Business Objectives *Background* *Business Objectives* *Business Success Criteria* **Assess Situation** *Inventory of Resources* *Requirements, Assumptions, and Constraints* *Risks and Contingencies* *Terminology* *Costs and Benefits* **Determine Data Mining Goals** *Data Mining Goals* *Data Mining Success Criteria* **Produce Project Plan** *Project Plan* *Initial Assessment of Tools and Techniques*	**Collect Initial Data** *Initial Data Collection Report* **Describe Data** *Data Description Report* **Explore Data** *Data Exploration Report* **Verify Data Quality** *Data Quality Report*	**Select Data** *Rationale for Inclusion/ Exclusion* **Clean Data** *Data Cleaning Report* **Construct Data** *Derived Attributes* *Generated Records* **Integrate Data** *Merged Data* **Format Data** *Reformatted Data* *Dataset* *Dataset Description*	**Select Modeling Techniques** *Modeling Technique* *Modeling Assumptions* **Generate Test Design** *Test Design* **Build Model** *Parameter Settings* *Models* *Model Descriptions* **Assess Model** *Model Assessment* *Revised Parameter Settings*	**Evaluate Results** *Assessment of Data Mining Results w.r.t. Business Success Criteria* *Approved Models* **Review Process** *Review of Process* **Determine Next Steps** *List of Possible Actions* *Decision*	**Plan Deployment** *Deployment Plan* **Plan Monitoring and Maintenance** *Monitoring and Maintenance Plan* **Produce Final Report** *Final Report* *Final Presentation* **Review Project** *Experience Documentation*

A bit more agile version




Business analysis
1
Analysis phase
2
Data exploration
Is developing model feasible?
3
Select algorithm
4
Data pipeline and feature engeneering
Development phase
5
Model building
6
Model evaluation
7
Presentation of results
8
Deployment plan
Deployment phase
9
Model operationalization
10
Model monitoring
Iteration s


Model experimentation and development
Data source
Data preparation
Modeling
Source code
Source repo
CI/CD
Pipeline building and testing
Packages
CD Pipeline Deployment
Featur e store
Model update cycle
Data preparation
Modeling
Model repo
Model serving
Continuous Deployment (CD)
Data validation
Data preparation
Model serving
Scaling
Logging
Monitoring
Trigger
Model monitoring
Model API
Live data

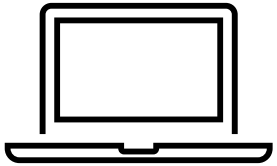
Tech stacks

Local

Python

Local file system

Component + artefact: dedicated folder

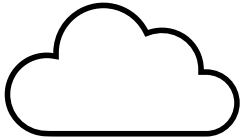


Local + cloud

Python
MLFlow, DVC, WandB

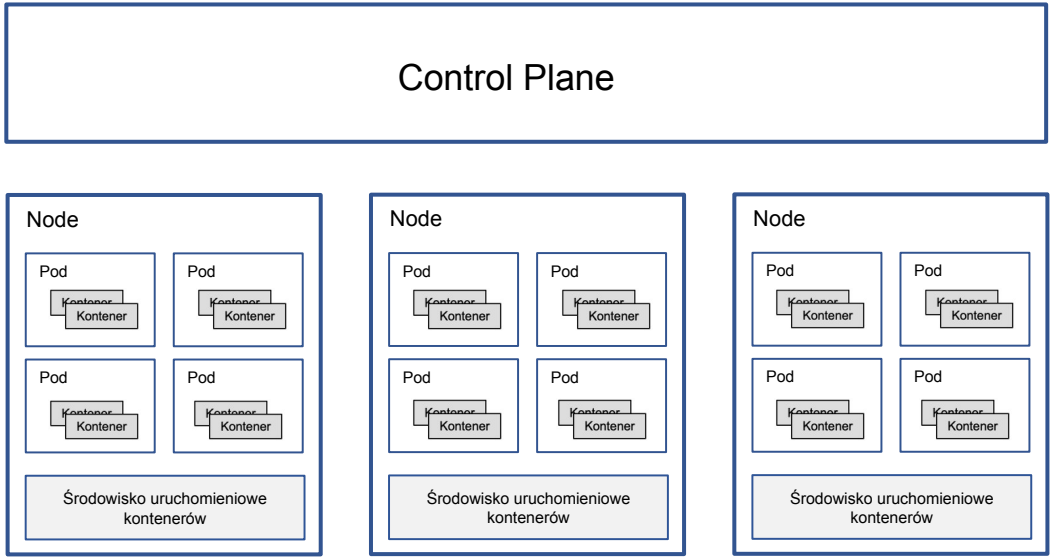
File system: local and/or cloud

Component + artefact: dedicated folder



Cluster

Python
MLFlow, DVC, WandB
Kubernetes, Kubeflow



ML / AI Infrastructure

DATA PREPARATION

Data Exploration & Processing
Paxata, TRIFACTA, alteryx, databricks, Qubole, HIVE

Data Version Control
Pachyderm, data iku

Feature Engineering and Storage
FEATURE LABS, Featuretools

Data Labeling
scale, Labelbox, figure eight, Amazon SageMaker, Google AI, Azure Machine Learning

Data Quality Checks
great_expectations

MODEL BUILDING

Hosted Notebooks Management
Amazon SageMaker, Azure Machine Learning, databricks, CLOUDERA, CO, DOMINO, Deepnote

Model Management, Version Tracking and Storage
databricks, Amazon SageMaker, Google AI, Azure Machine Learning, ALGORITHMIA, DOMINO, data iku

Experiment Tracking
Weights & Biases, comet, TensorBoard, mlflow

Model Optimization Hyper Parameter
SIGOPT, Weights & Biases, Amazon SageMaker, anyscale

Auto ML
H2O.ai, DataRobot, Amazon SageMaker, Google AI, Azure Machine Learning, data iku

Model Training
Amazon SageMaker, Azure Machine Learning, Google AI, Kubeflow, anyscale

Model Evaluation
TensorBoard

Model Explainability
fiddler, TensorBoard

PRODUCTION

Model Observability
arize, DataRobot, Other startups

Model Compliance & Audit
fiddler, Other startups

Model Deployment and Serving
Amazon SageMaker, Kubeflow, ALGORITHMIA, DataRobot, Google AI, Azure Machine Learning

Model Validation
arize, fiddler, Other startups

Platform Specific Model Builds
OctoML

Machine Learning tools & platforms landscape - v.1.0 January 2021

Presented by MLReef

DATA MANAGEMENT

Data Exploration & Management

COHESITY, rubrik, allegro.ai, ALLUXIO, Amundsen, druid, nuodi, MLReef, databricks, ALGORITHMIA, Spark, APARAVI, ATSCALE, amazon account, CAZENA, Google Cloud, CLOUDERA, kaggle, datagrok, data iku, ClearSky, Data Lake, DATERA, dremio, elastifile, erwin, Excelero, Fluree, GRAN, HYCU, imply, komprise, Parquet, pilosa, presto, tanx, VEARCH, vexata, YellowBrick

Data Labelling

appen, Dataturks, doccano, Dataiku, iMerit, prodigy, Labelbox, SUPERVISELY, Playment, scale, snorkel, HIVE

Data Streaming

ALLUXIO, nuodi, kafka, CONFLUENT, VALOHAI, Google Cloud, striim

Data Version Control

databricks, data iku, DVC, FLOYDHUB, MLReef, Pachyderm, Waterline Data, QRI, allegro.ai

Data Generation

scale, scrapinghub, DATPROF

Data Privacy

airlock, Celantur, mostly, PySyft, TUMULT

Data Quality Checks

arize, great_expectations, Naveego

MODELLING

Notebook / Code Management

colab, databricks, data iku, Deepnote, DOMINO, FLOYDHUB, Google Cloud, kaggle, MLReef, OpenML, polyaxon, Pachyderm, Weights & Biases, alteryx

Data Processing & Visualization

alteryx, ASCENDLIO, colab, DASK, data iku, databricks, dotData, Flyte, gluent, iguazio, imply, incorta, mlflow, MLReef, MODIN, Naveego, OpenML, Pachyderm, pilosa, presto, Prometheus, SAS, snorkel, SQLFlow, Starburst, vaex, VALOHAI, allegro.ai, Weights & Biases

Feature Engineering

dotData, FEAST, Featuretools, Pachyderm, tecton, SAS, data iku, iguazio

Model Training

alteryx, iguazio, Google Cloud, colab, databricks, data iku, DOMINO, dotscience, FLOYDHUB, Flyte, ludwig, kaggle, MLReef, mlflow, PerceptiLabs, snorkel, VALOHAI, SAS, Anyscale, Pachyderm, OpenML

Experiment Tracking

iguazio, allegro.ai, comet, data iku, DataRobot, datmo, DOMINO, FLOYDHUB, ludwig, mlflow, MLReef, OpenML, polyaxon, SPELL, VALOHAI, Google Cloud, Weights & Biases, Losswise

Model (Hyperparameter) Optimization

Amazon SageMaker, Angel, comet, DataRobot, polyaxon, SIGOPT, SPELL, tune, OPTUNA, alteryx, talos, allegro.ai, SAS

Auto ML

DataRobot, Determined AI, dotData, Google Cloud, kubeflow, TPOT, TransmogrifAI, iguazio, Amazon SageMaker

Model Management

alteryx, Amazon SageMaker, ALGORITHMIA, allegro.ai, databricks, data iku, Determined AI, dockerhub, DOMINO, dotData, FLOYDHUB, LUCIO, Google Cloud, OpenML, kubeflow, iguazio, MLReef, modzy, mlflow, PerceptiLabs, polyaxon, SAS, TUMULT, VALOHAI, Verta

Model Evaluation

arize, TensorBoard, MLPerf, Streamlit

Model Explainability

fiddler, InterpretML, LUCIO, PerceptiLabs, Shap, TensorBoard, Amazon SageMaker

Frameworks & major libraries

Chainer, mxnet, Keras, Spark MLlib, Caffe2, PyTorch, ONNX, TensorFlow, spaCy, XGBoost, CNTK, theano, matplotlib, fastai, julia, torch

CONTINUOUS DEPLOYMENT

Data Flow Management

ALLUXIO, Spark, ASCENDLIO, kafka, data iku, dotData, HYCU, PREFECT

Feature Transformation

FEAST, Featuretools, tecton, iguazio, data iku

Monitoring

ALGORITHMIA, arize, Google Cloud, data iku, DATADOG, DataRobot, DOMINO, iguazio, Losswise, snorkel, unravel, VALOHAI, Verta, kubeflow, datatron, Amazon SageMaker, alteryx

Model Compliance & Audit

ALGORITHMIA, SAS, kubeflow

Model Deployment & Serving

AIBLE, ALGORITHMIA, allegro.ai, cortex, data iku, datatron, datmo, DOMINO, dotData, FLOYDHUB, FRITZ AI, iguazio, Kubeflow, mlflow, modzy, OpenML, PaperSpace, PredictionIO, SAS, SELDON, SPELL, Streamlit, Amazon SageMaker, VALOHAI, Verta, kubeflow, Google Cloud, Azure, Amazon SageMaker, alteryx

Model Validation

arize, datatron, fiddler, LUCIO, MLPerf, SAS, Streamlit

Model Compatibility

MMdnn, ONNX, OpenDVL

COMPUTING MANAGEMENT

Computing & Data Infrastructure

Google Cloud, Azure, Amazon SageMaker, CLOUDERA, ORACLE, Amazon EC2, linode, DigitalOcean, Linux Foundation, Docker

Environment Management

CONDA, databricks, datmo, MAHOUT, MLReef, allegro.ai

Resource Allocation

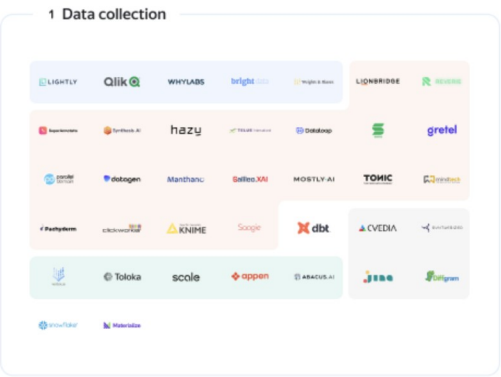
ALGORITHMIA, Google Cloud, MLReef, Azure, databricks, data iku, Determined AI, FLOYDHUB, Amazon SageMaker, polyaxon, SPELL, VALOHAI, allegro.ai

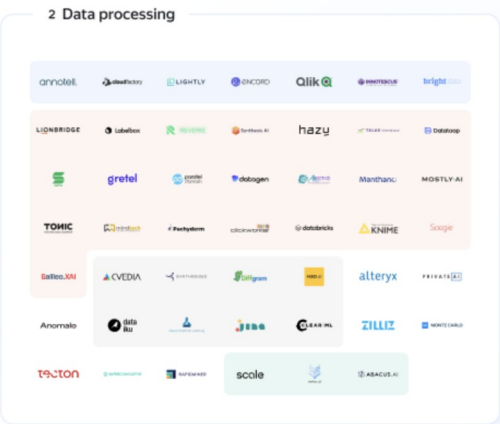
SCALING

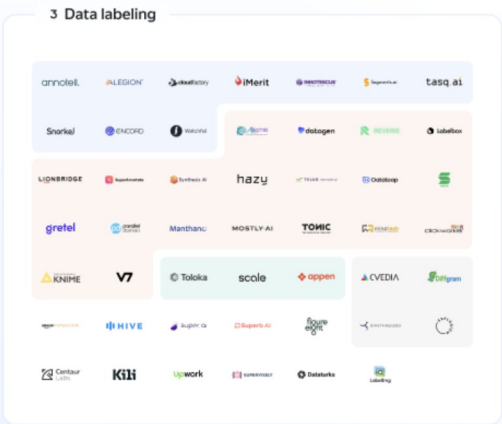
orgo, Azure, DATADOG, datatron, datmo, Google Cloud, Amazon EC2, SELDON, tvm, kubeflow

SECURITY & PRIVACY

Azure, DATADOG, modzy, PySyft, TUMULT, ALGORITHMIA







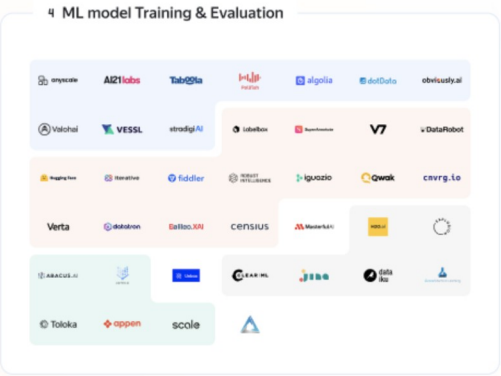
THE SEQUENCE

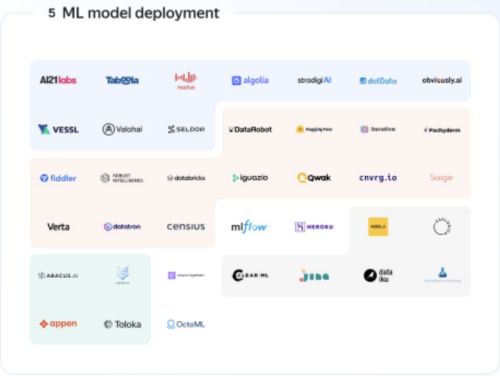
ML Value Chain Landscape

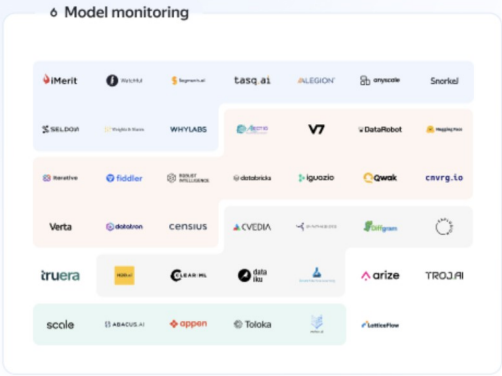
Created by your ML peers. Powered by TheSequence.

Vendors represented in more than 1 stage are indicated by color coded frames:

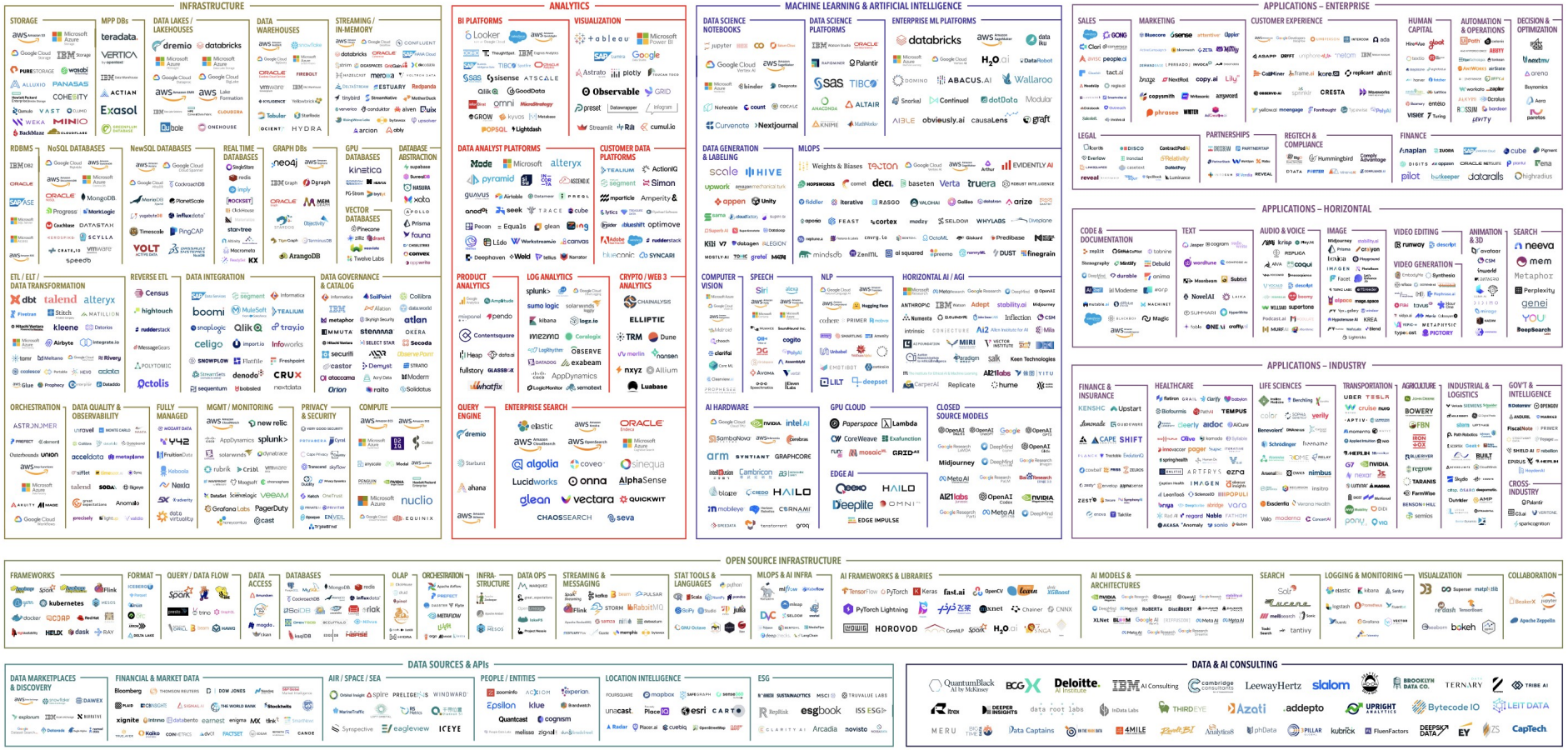
2 for the vendors covering **2** stages	3 for the vendors covering **3** stages	4 for the vendors covering **4** stages	5 for the vendors covering **5** stages







THE 2023 MAD (MACHINE LEARNING, ARTIFICIAL INTELLIGENCE & DATA) LANDSCAPE



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FIRSTMARK
EARLY STAGE VENTURE CAPITAL